

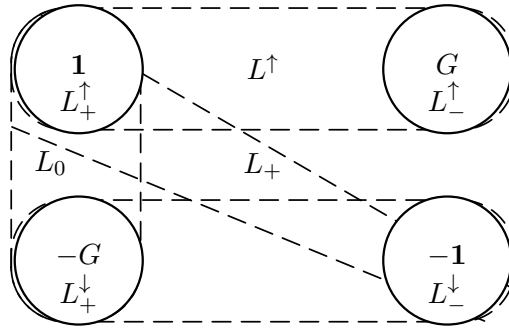


FIGURE 1-1. Connectivity properties of the Lorentz group, L , and its subgroups: the proper Lorentz group, L_+ ; the orthochronous Lorentz group, $L^\uparrow$; the orthochorous Lorentz group, L_0 ; and the restricted Lorentz group, $L_+^\uparrow$.

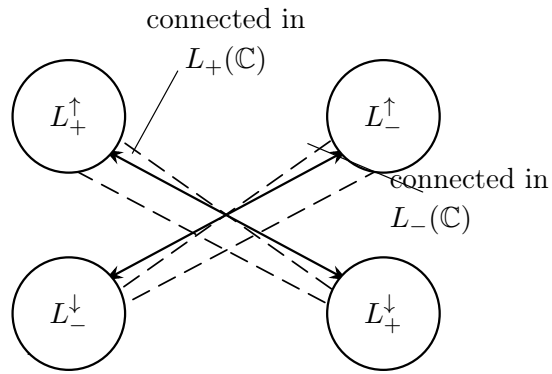


FIGURE 1-2. Connectivity properties of the complex Lorentz group, $L(\mathbb{C})$. There are two connected components: $L_+(\mathbb{C})$, the proper complex Lorentz group, which contains $L_+^\uparrow$ and $L_+^\downarrow$; and $L_-(\mathbb{C})$, which contains $L_-^\uparrow$ and $L_-^\downarrow$.

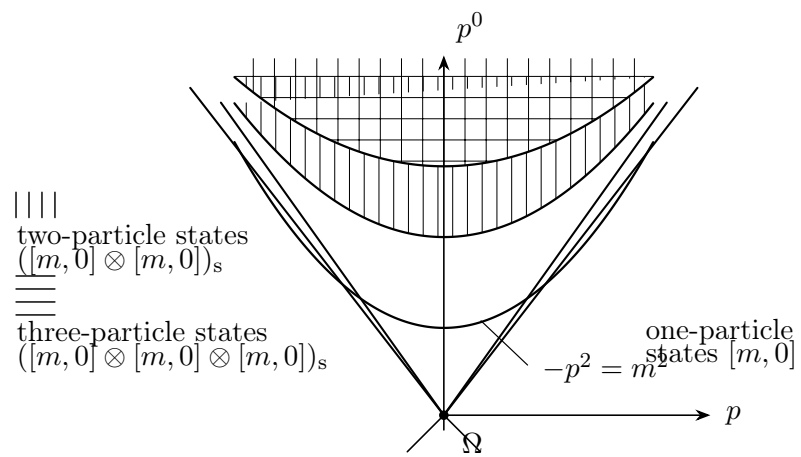


FIGURE 1-3. The spectrum of a theory of neutral scalar mesons of mass m without bound states.

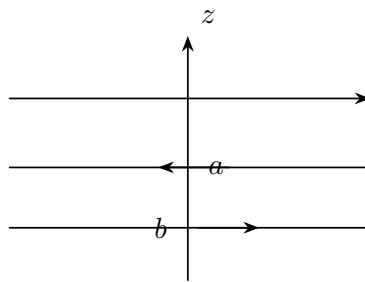


FIGURE 2-1. The contour C in the plane of $z = \xi - i\eta$.

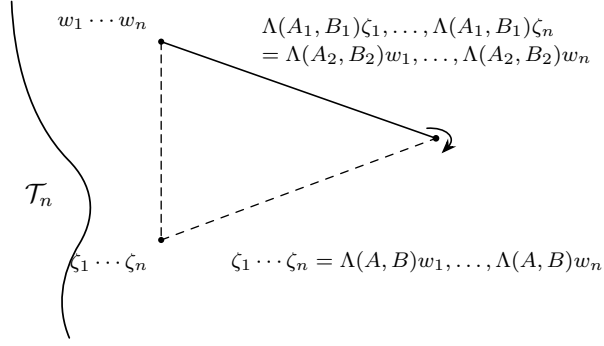


FIGURE 2-2. The single-valuedness of the functions as defined throughout the extended tube $\mathcal{T}'_n$ follows from the fact that, if $w_1, \dots, w_n$ and $\Lambda w_1, \dots, \Lambda w_n \in \mathcal{T}_n$, then there is a curve of proper complex Lorentz transformations $\{\Lambda(t); 0 \leq t \leq 1\}$ with $\Lambda(0) = \mathbf{1}$, $\Lambda(1) = \Lambda$, such that $\Lambda(t)w_1, \dots, \Lambda(t)w_n \in \mathcal{T}_n$ for $0 \leq t \leq 1$. Here $A = A_1^{-1}A_2$, $B = B_1^{-1}B_2$.

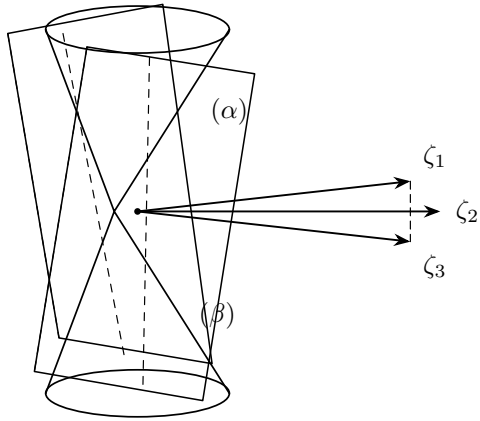


FIGURE 2-3. The cone of space-like vectors corresponding to a Jost point $\zeta_1, \zeta_2, \zeta_3$. The planes (α) and (β) separate the cone spanned by $\zeta_1, \zeta_2, \zeta_3$ from the interior of the plus and minus light cones.

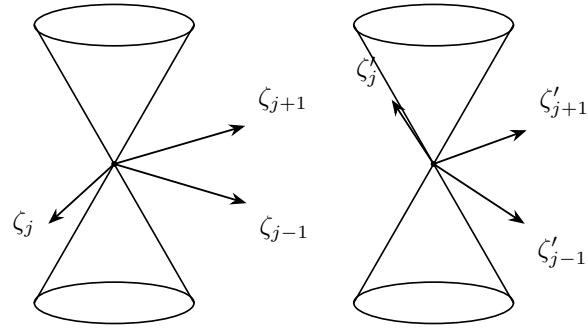


FIGURE 2.4. A configuration of vectors $\zeta_{j-1}, \zeta_j, \zeta_{j+1}$ appearing for a Jost point of $\mathcal{T}'_n$ and $P(j, j+1)\mathcal{T}'_n$. The ζ' 's are obtained from the ζ 's by $\zeta'_{j-1} = \zeta_{j-1} + \zeta_j$, $\zeta'_j = -\zeta_j$, $\zeta'_{j+1} = \zeta_{j+1} + \zeta_j$.

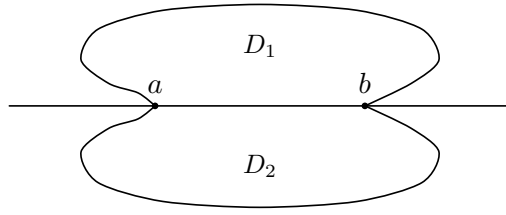


FIGURE 2-5. Domains D_1 and D_2 , of definition of F_1 and F_2 , respectively.

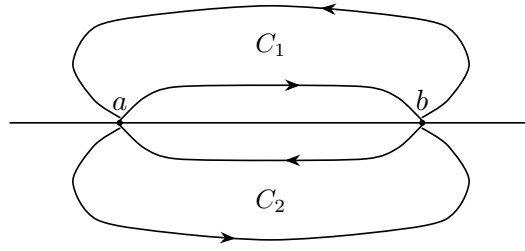


FIGURE 2-6. The contours C_1 and C_2 .

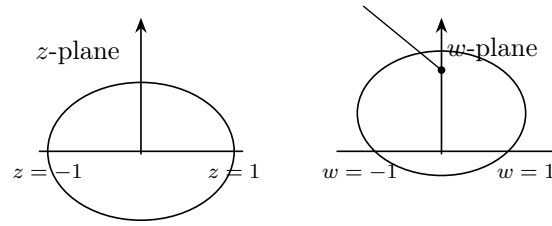


FIGURE 2-7. Illustration of the mapping $w = (u + z)/(1 + uz)$. As z runs over the unit circle $|z| = 1$, w runs over the circle indicated. The points $z = \pm 1$ always are mapped into $w = \pm 1$.

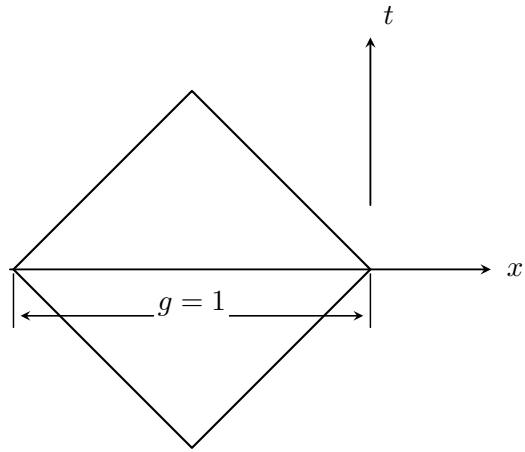


Figure A.1. The dependence domain of the region where $g = 1$.

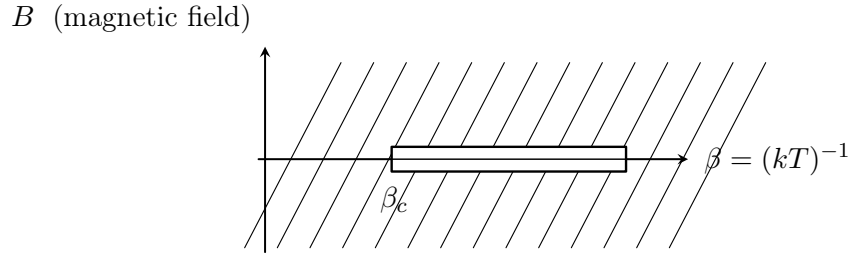
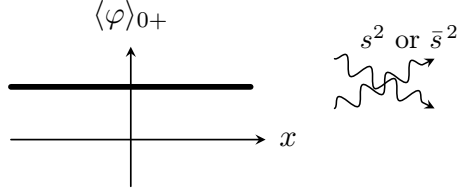
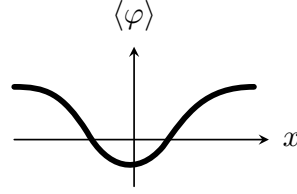


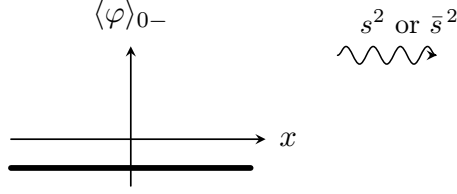
Figure A.2. Phase diagram of the Ising ferromagnet (nearest-neighbor interaction). The phase is unique and the correlation functions decay exponentially for all values of β and B except those satisfying $B = 0$, $\beta_c \leq \beta < \infty$, where there are exactly two phases. The phase diagram for the $\lambda\varphi^4 + \mu\varphi^2 + \nu\varphi$ model is similar, with B replaced by ν and β by λ . The solution is known to be unique for $\nu \neq 0$ and for all sufficiently small positive λ . It is known that at least two solutions, I and II, exist for all sufficiently large λ with $\varphi \rightarrow -\varphi$ symmetry broken in the sense that $\langle \Omega_{0I} | \varphi_1(x) | \Omega_{0I} \rangle = -\langle \Omega_{0II} | \varphi_1(x) | \Omega_{0II} \rangle \neq 0$.



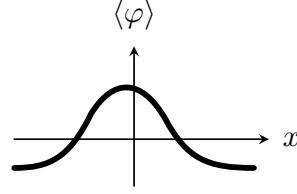
Ψ_{0+}
 $\mathcal{H}_{0+}$ In the + vacuum, Ψ_{0+} , the expectation value $\langle \varphi(x) \rangle_{0+}$ of $\varphi(x)$ is positive and constant.



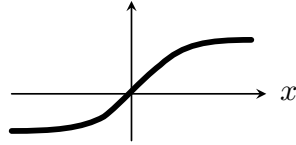
A two-soliton state of $\mathcal{H}_{0+}$; it has the same expectation value of $\varphi(x)$ as Ψ_{0+} for x far to the left and right.



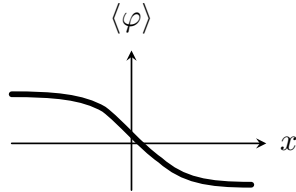
Ψ_{0-}
 $\mathcal{H}_{0-}$ In the - vacuum, Ψ_{0-} , the expectation value $\langle \varphi(x) \rangle_{0-}$ of $\varphi(x)$ is negative and constant.



A two-soliton state of $\mathcal{H}_{0-}$; it has the same expectation value of $\varphi(x)$ as Ψ_{0-} for x far to the left and right.



$\mathcal{H}_s$ This state is obtained from Ψ_{0-} by applying the automorphism s and as a result changing the asymptotic value of the expectation value of $\varphi(x)$ far to the right. It is also obtained by applying $\bar{s}$ to Ψ_{0+} .



$\mathcal{H}_{\bar{s}}$ This state is obtained from Ψ_{0+} by applying the automorphism, $\bar{s}$, and therefore changing the asymptotic value of the expectation value of $\varphi(x)$ far to the right. It is also obtained by applying s to Ψ_{0-} .

Figure A.3. Behavior of $\langle \varphi(x) \rangle$ as a function of space coordinate for some typical states of $\mathcal{P}(\varphi^4)_2$ in the two-phase region. The action of the automorphisms $s^2 = s \circ s$, $\bar{s}^2 = \bar{s} \circ \bar{s}$ and $s \circ \bar{s}$ are indicated by the wavy arrows. In particular, note that the action of the composite automorphism $s \circ \bar{s}$ is to change the asymptotic value of the expectation value of $\varphi(x)$ both to the right and left.